

Centre No.						Paper Reference						Surname	Initial(s)	
Candidate No.						6	6	7	7	/	0	1	Signature	

Paper Reference(s)

6677/01

Edexcel GCE

Mechanics M1

Advanced/Advanced Subsidiary

Wednesday 12 January 2005 – Afternoon

Time: 1 hour 30 minutes

Examiner's use only

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Team Leader's use only

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[illegible]

Materials required for examination

Items included with question papers

Mathematical Formulae (Lilac or Green)	Nil
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Candidates may use any calculator EXCEPT those with facility for symbolic algebra, differentiation and/or integration. Thus candidates may NOT use calculators such as the Texas Instruments T1 89, T1 92, Casio CFX 9970G, Hewlett Packard HP 48G.

Instructions to Candidates

In the boxes above, write your centre number, candidate number, your surname, initial(s) and signature.

You must write your answer to each question in the space following the question.

If you need more space to complete your answer to any question, use additional answer sheets.

Whenever a numerical value of g is required, take $g = 9.8 \text{ ms}^{-2}$.

When a calculator is used, the answer should be given to an appropriate degree of accuracy.

Information for Candidates

A booklet 'Mathematical Formulae and Statistical Tables' is provided.

Full marks may be obtained for answers to ALL questions.

This paper has seven questions. There are no blank pages.

The total mark for this paper is 75.

Advice to Candidates

You must ensure that your answers to parts of questions are clearly labelled.

You must show sufficient working to make your methods clear to the examiner. Answers without working may gain no credit.

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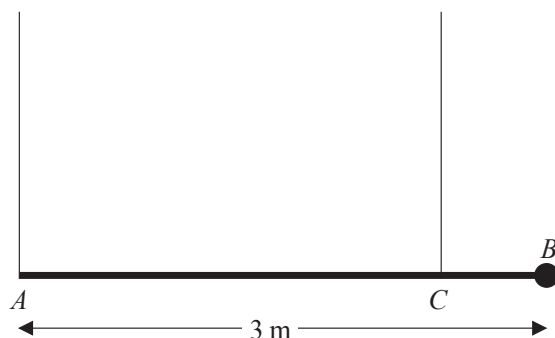
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1. A particle P of mass 1.5 kg is moving along a straight horizontal line with speed 3 m s^{-1} . Another particle Q of mass 2.5 kg is moving, in the opposite direction, along the same straight line with speed 4 m s^{-1} . The particles collide. Immediately after the collision the direction of motion of P is reversed and its speed is 2.5 m s^{-1} .
- (a) Calculate the speed of Q immediately after the impact. (3)
- (b) State whether or not the direction of motion of Q is changed by the collision. (1)
- (c) Calculate the magnitude of the impulse exerted by Q on P , giving the units of your answer. (3)

Figure 1



(a) the tension in the rope at C ,

(b) the distance CB . (5)

[illegible]

3.

A speed-time graph for a car. The vertical axis is labeled 'speed (m s^{-1})' and the horizontal axis is labeled 'time (s)'. The origin is marked O . The graph consists of three segments: a straight line from $(0,0)$ to $(4,9)$, a horizontal line at 9 m s^{-1} from $t=4$ to $t=20$, and a straight line from $(20,9)$ to $(25,u)$. Dashed lines indicate the coordinates of these points.

(a) the distance covered by the sprinter in the first 20 s of the race,

(2)

(b) the value of u ,

(4)

(c) the deceleration of the sprinter in the last 5 s of the race.

(3)

[illegible]

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Question 3 continued

Q3

(Total 9 marks)

4.

A diagram showing a particle P on an inclined plane. The plane is at an angle of 20° to the horizontal. A force of magnitude XN acts on the particle, directed up the plane.

(a) the normal reaction of the plane on P ,

(b) the value of X .

The force of magnitude X newtons is now removed.

(c) Show that P remains in equilibrium on the plane.

(4)

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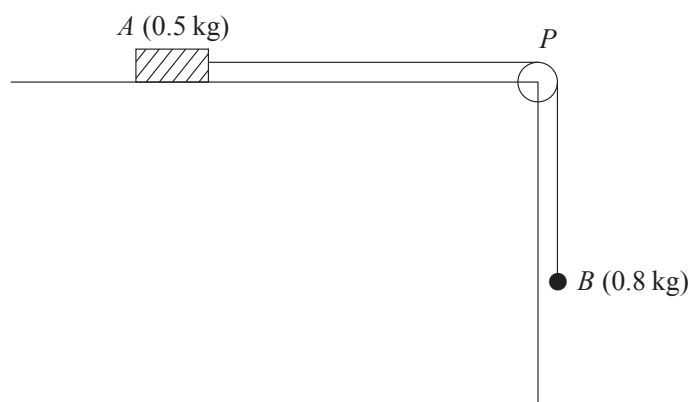
(Total 10 marks)

Q4

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5.

Figure 4



A block of wood A of mass 0.5 kg rests on a rough horizontal table and is attached to one end of a light inextensible string. The string passes over a small smooth pulley P fixed at the edge of the table. The other end of the string is attached to a ball B of mass 0.8 kg which hangs freely below the pulley, as shown in Figure 4. The coefficient of friction between A and the table is μ . The system is released from rest with the string taut. After release, B descends a distance of 0.4 m in 0.5 s. Modelling A and B as particles, calculate

- (a) the acceleration of B , (3)
- (b) the tension in the string, (4)
- (c) the value of μ . (5)
- (d) State how in your calculations you have used the information that the string is inextensible. (1)

(Total 13 marks)

Q5

- (a) the deceleration of S ,

(2)

- (b) the speed of S at C .

(3)

- (c) Show that the mass of S is 0.1 kg.

(2)

(d) Calculate the time between the instant that S rebounds from the wall and the instant that S comes to rest.

(6)

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(Total 13 marks)

Q6

7. Two ships P and Q are travelling at night with constant velocities. At midnight, P is at the point with position vector $(20\mathbf{i} + 10\mathbf{j})$ km relative to a fixed origin O . At the same time, Q is at the point with position vector $(14\mathbf{i} - 6\mathbf{j})$ km. Three hours later, P is at the point with position vector $(29\mathbf{i} + 34\mathbf{j})$ km. The ship Q travels with velocity $12\mathbf{j}$ km h⁻¹. At time t hours after midnight, the position vectors of P and Q are $\mathbf{p}$ km and $\mathbf{q}$ km respectively. Find

- (2)

- (4)

(c) By finding an expression for $\overrightarrow{PQ}$, show that

(5)

(d) find the time, to the nearest minute, at which the lights on Q move out of sight of the observer.

(5)

